SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	**Assessment:**	Comparative Analysis
Weightage:	30%	**Total Marks:**	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	2023-2027	**Date:**	26-08-2026

Instructions to Students

• Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
• Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption. • Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
• Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable. • Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

QUESTIONS

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.

2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.

3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.

4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.

5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

Code of Conduct

I ____________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with	Shows reasonable	Basic interpretation;	No meaningful interpretation

		clear interpretation and reasoning	interpretation with minor gaps	lacks depth	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Marks Evaluation:

Q. No	Scope of Items **(20)**	Comparison Depth **(25)** Use of Evidence **(20)** Critical Thinking **(20)** Clarity of Presentation **(15)**	**Total (out of 100)**
1	/ 4	/ 4 / 4 / 4 / 4	
2	/ 4	/ 4 / 4 / 4 / 4	

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Assessment Tool 2 — Comparative Analysis

Model Answer Key • CO4 (BL4 — Precision)

Course Code	CSA12	**Course Title**	Computer Architecture

CO Assessed	CO4 — Precision (BL4)	**Weightage** 30% · 100 Marks

CO4 Statement. Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory

with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.

*Each answer is written in comparative form per the rubric — **scope, comparison depth, evidence (typical figures), critical thinking, and clear structure**. Latency/bandwidth figures are representative order-of-magnitude values and vary by process node and generation.*

Scope. Compare the three on-chip SRAM cache levels by access latency, bandwidth and throughput, and analyse how each shapes processor performance and memory-access efficiency.

Comparison across cache levels

Typical size	32–64 KB (split I/D) 256 KB – 1 MB	4–32 MB
Access latency	~1 ns (~4 cycles) ~3–5 ns (~10–14 cyc)	~10–20 ns (~30–50 cyc)
Bandwidth	Highest (≥2 accesses/cyc) High	Moderate–high
Placement	Private per core Private / cluster	Shared across cores
Technology	SRAM SRAM	SRAM

Typical hit rate	~90–95 % ~ high (local)	captures multi-core reuse
Primary role	Feed pipeline instantly Catch L1 spillover	Shared victim / coherence

Each step down the hierarchy trades **latency and bandwidth for capacity**: L1 is smallest but fastest and sustains the highest throughput (it must serve loads/stores almost every cycle); L3 is largest but slowest, absorbing misses from many cores before the ~100 ns trip to DRAM.

Evidence — effect on processor performance (AMAT)

Average Memory Access Time with a three-level hierarchy

$AMAT = t_{L1} + m_{L1}(t_{L2} + m_{L2}(t_{L3} + m_{L3} \cdot t_{DRAM}))$

With t_{L1}=1, m_{L1}=0.05, t_{L2}=4, m_{L2}=0.15, t_{L3}=15, m_{L3}=0.25, t_{DRAM}=100 ns:
$AMAT = 1 + 0.05(4 + 0.15(15 + 0.25\times100)) = 1 + 0.05\times10 = \mathbf{1.50\ ns}$

Versus no cache (every access = 100 ns): speedup = 100 / 1.50 ≈ **67×**.

→ Multi-level caching converts a 100 ns memory into an effective ~1.5 ns memory.

Critical analysis

• **Locality is the enabler:** temporal and spatial locality let small fast levels capture most accesses, so the common case runs at L1 speed while rare misses fall back to larger levels. • **Latency vs capacity trade-off:** making L1 larger would slow it down; the tiered design keeps each level optimally sized for its access-time target. • **Multicore role of L3:** a shared LLC improves inter-core data sharing and cache-coherence efficiency, and raises the effective hit rate for multi-threaded workloads.

Key comparative takeaway

L1 optimises **latency**, L3 optimises **capacity/sharing**, and L2 bridges them.
Together they hide the processor-memory gap, turning ~100 ns DRAM into ~1.5 ns effective access — the single biggest contributor to sustained instruction throughput.

Scope. Compare SRAM and DRAM by latency, bandwidth and throughput, and justify why SRAM is used for cache and DRAM for main memory.

Head-to-head comparison

Cell structure	6-transistor bistable latch 1 transistor + 1 capacitor
Access latency	~0.5–2 ns ~50–100 ns (RAS/CAS + activate)
Bandwidth	Very high (wide on-chip) High off-chip (DDR5 ~ up to 51 GB/s/ch)
Density	Low (large cell) High (small cell)
Refresh	None (static) Periodic refresh needed (dynamic)
Cost per bit	High Low
Power	High static leakage Lower static + refresh power
Volatility	Volatile Volatile

Justification of roles (use of evidence)

- **SRAM for cache:** caches demand the **lowest, most deterministic latency** and are small (KB–MB), so SRAM's ~1 ns access and refresh-free operation are worth its high cost/area. Its speed matches the pipeline; density is not the priority at these sizes.
- **DRAM for main memory:** main memory needs **large capacity at low cost/bit** (GBs). DRAM's 1T1C cell is dense and cheap, and its ~80–100 ns latency is acceptable because caches already absorb most accesses. Refresh overhead is a fair price for the capacity.

Why not build everything from one technology

All-SRAM main memory: ~1 ns but 6T cell → ~6–8× the area and far higher $/bit and leakage → GBs infeasible.

All-DRAM caches: dense/cheap but ~50–100 ns → would stall the pipeline every access, collapsing IPC. → The hierarchy pairs each technology with the tier where its strength dominates.

Critical analysis

• The comparison is fundamentally **speed vs density vs cost**: SRAM wins speed, DRAM wins density and cost. Neither is 'better' — suitability depends on the tier's requirement. • DRAM's refresh and higher latency are hidden behind SRAM caches, so the system enjoys SRAM-like effective latency with DRAM-like capacity.

Key comparative takeaway

Use **SRAM where latency dominates** (small, fast caches) and **DRAM where capacity and cost dominate** (large main memory).

The pairing gives the processor both fast access (via cache) and large working memory (via DRAM) without paying SRAM prices for GBs or DRAM latency for every access.

Scope. Examine these three memory-management mechanisms by access latency, memory utilisation and system throughput, and analyse how they jointly deliver efficient memory management.

Comparison of mechanisms

Managed by	Hardware (transparent)	OS + MMU	OS + MMU/TLB
Purpose	Cut CPU–DRAM latency	Large address space + isolation	Map & fetch pages on demand
Granularity	Cache line (~64 B)	Address space	Page (4 KB / huge 2 MB)
Latency added	~1–20 ns on hit	Translation overhead	TLB ~1 ns; fault µs–ms
Key benefit	Speed (locality)	Capacity beyond RAM	Efficient RAM utilisation
Main risk	Miss → DRAM	—	Page fault / thrashing

The **TLB** is itself a cache — of virtual-to-physical translations — which links the two ideas: cache principles (locality, fast lookup) are reused to make paging affordable.

Evidence — quantifying the overheads

Translation + page-fault effect on effective access time

TLB: EAT = $h(t_{tlb}+t_{mem}) + (1-h)(t_{tlb}+t_{walk}+t_{mem})$

$h=0.98$, $t_{tlb}=1$, $t_{walk}=100$, $t_{mem}=100$ ns → 0.98×101 + 0.02×201 = **103 ns**

Page faults: EAT = $(1-p)\cdot t_{mem} + p\cdot t_{fault}$, $t_{fault}=100$ µs (SSD)
$p=0.001$ → 0.999×100 + 0.001×100000 = **199.9 ns** (≈2× slower from just 0.1 % faults).

Critical analysis

- **Latency hierarchy:** cache (ns) ≪ TLB-miss walk (×100 ns) ≪ page fault (µs–ms). Managing memory well means keeping accesses as high in this hierarchy as possible.
- **Memory utilisation:** paging loads only needed pages, supports over-commit and multiprogramming, and eliminates external fragmentation — raising RAM utilisation and CPU throughput.
- **Throughput risk — thrashing:** if the combined working set exceeds RAM, page-fault traffic dominates and throughput collapses; huge pages and adequate RAM mitigate this.

Key comparative takeaway

Cache attacks the **ns** CPU-memory gap; virtual memory/paging manage the **µs–ms** capacity gap and enable multiprogramming.

Used together — with the TLB caching translations — they give fast access, large usable memory and high utilisation, provided faults are kept rare to avoid thrashing.

Scope. Compare NAND Flash and DRAM by latency, bandwidth and throughput, and evaluate which tier of the memory hierarchy each suits.

Head-to-head comparison

Access latency	~50–100 ns	Read ~25–100 µs; erase ~ms
Bandwidth	High (tens of GB/s)	Moderate (GB/s via parallel dies)
Addressability	Byte-addressable	Page/block (not byte)
Volatility	Volatile	Non-volatile
Capacity / cost	GBs, moderate $/bit	GB–TB, low $/bit
Endurance	Effectively unlimited	Limited P/E cycles (SLC>…>QLC)
Best tier	Main memory	Secondary storage (SSD) / swap

Throughput analysis (use of evidence)

Latency vs throughput — the parallelism trick

Per-operation latency gap is ~1000× (DRAM ~100 ns vs NAND ~100 µs read).

Yet NAND SSDs reach multi-GB/s **sequential** throughput by striping across many dies/channels, hiding per-op latency — good for bulk transfers, poor for fine-grained random low-latency access. DRAM delivers both low latency AND high random-access throughput, which NAND cannot match.

Suitability & justification

- **DRAM → main memory:** byte-addressable, ~100 ns, unlimited endurance → ideal for the working set the CPU accesses directly through the caches.
- **NAND Flash → secondary storage / fast swap:** non-volatile, dense and cheap → ideal for persistent bulk storage; its µs latency rules it out as main memory but suits an SSD tier and as a fast paging/swap backing store bridging DRAM and disk.

Critical analysis

- The ~1000× latency gap and lack of byte addressability mean NAND **complements** rather than replaces DRAM; each fills a different tier.
- Endurance/wear (limited P/E cycles) requires wear-levelling and over-provisioning — a design cost DRAM does not carry; persistent-memory tiers (e.g., NVMe/SCM) are narrowing the gap.

Key comparative takeaway

DRAM = **fast, volatile, byte-addressable** main memory; NAND = **slower, non-volatile, dense** storage. The hierarchy uses DRAM for the active working set and NAND for persistent, high-capacity storage and swap — a latency/persistence/cost division of labour.

Scope. Compare MRAM and RRAM by access time, throughput, energy efficiency and endurance, and assess their roles in future hierarchical memory systems.

Head-to-head comparison

Storage principle	Magnetic tunnel junction	Resistive filament in oxide
Read access time	~10–40 ns	~10–100 ns
Write	Slower, higher energy	Moderate; some variability
Endurance	High (~10^{12}–10^{15})	Lower (~10^{6}–10^{9})
Energy	Low static; write-heavy	Low read energy
Density	Moderate	High (crossbar, 3D-stackable)
Volatility	Non-volatile	Non-volatile
Standout trait	Fast + very endurant	Dense + compute-in-memory

Evidence & throughput

• Both are **non-volatile**, so they draw ~0 retention power (no refresh, no leakage to hold data) — a major energy-efficiency advantage over SRAM/DRAM for idle/always-on systems. • **MRAM** read speed approaches SRAM/DRAM, giving good random throughput; its very high endurance lets it tolerate cache-like write rates. • **RRAM** crossbars enable massively parallel analog matrix-vector multiply (**compute-in-memory**), giving very high effective throughput for AI/neuromorphic workloads while avoiding data movement.

Future hierarchical-memory applications

- **MRAM:** non-volatile last-level cache / embedded NVM and 'normally-off' processors that resume instantly — a step toward a fast + persistent 'universal memory'.

- **RRAM:** a dense storage-class-memory tier between DRAM and NAND, and in-memory-computing accelerators for edge AI.

Advantages and limitations (critical thinking)

MRAM	Fast reads, huge endurance, non-volatile Lower density than DRAM/RRAM; write energy; cost; maturity
RRAM	High density, low read energy, enables CIM Lower endurance, write variability/reliability; maturity

Key comparative takeaway

MRAM leads on speed and endurance (→ persistent cache / universal memory); **RRAM** leads on density and in-memory compute (→ storage-class memory / AI accelerators).

Neither yet fully replaces SRAM/DRAM, but both enable non-volatile, energy-efficient tiers and 'normally-off' and near-memory computing in future hierarchies.